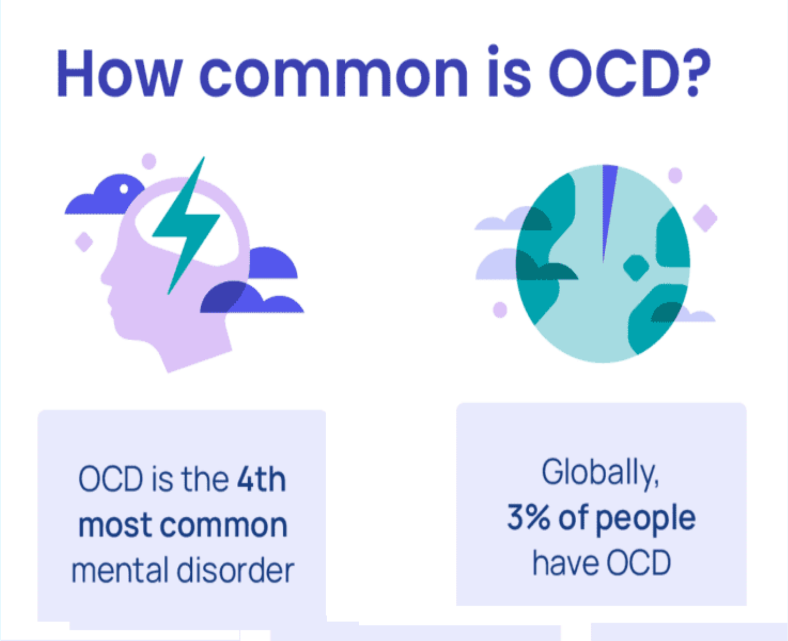
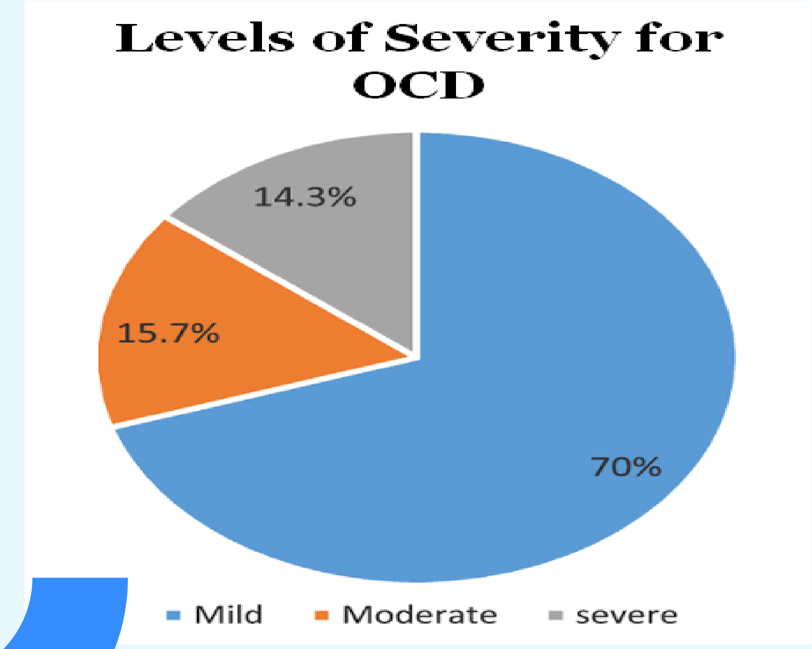
NEUROKAVACH

AI-Powered Wearable for Early Intervention in Obsessive-Compulsive Behaviour

Presented by **Team ZeroStack**

DO YOU KNOW?





According to the **World Health Organization** (**WHO**), OCD is the **4th most common** mental disorder in the world.

Globally, OCD affects an estimated **2-3% of the population**. This translates to roughly **170 million to 250 million people** worldwide living with the condition.

Understanding the rationale behind compulsions, particularly how **individuals respond to specific triggers**, is crucial in interpreting compulsive behaviors, which can potentially help **enhance compulsion detection and facilitate intervention delivery at the right timing.**

[…]

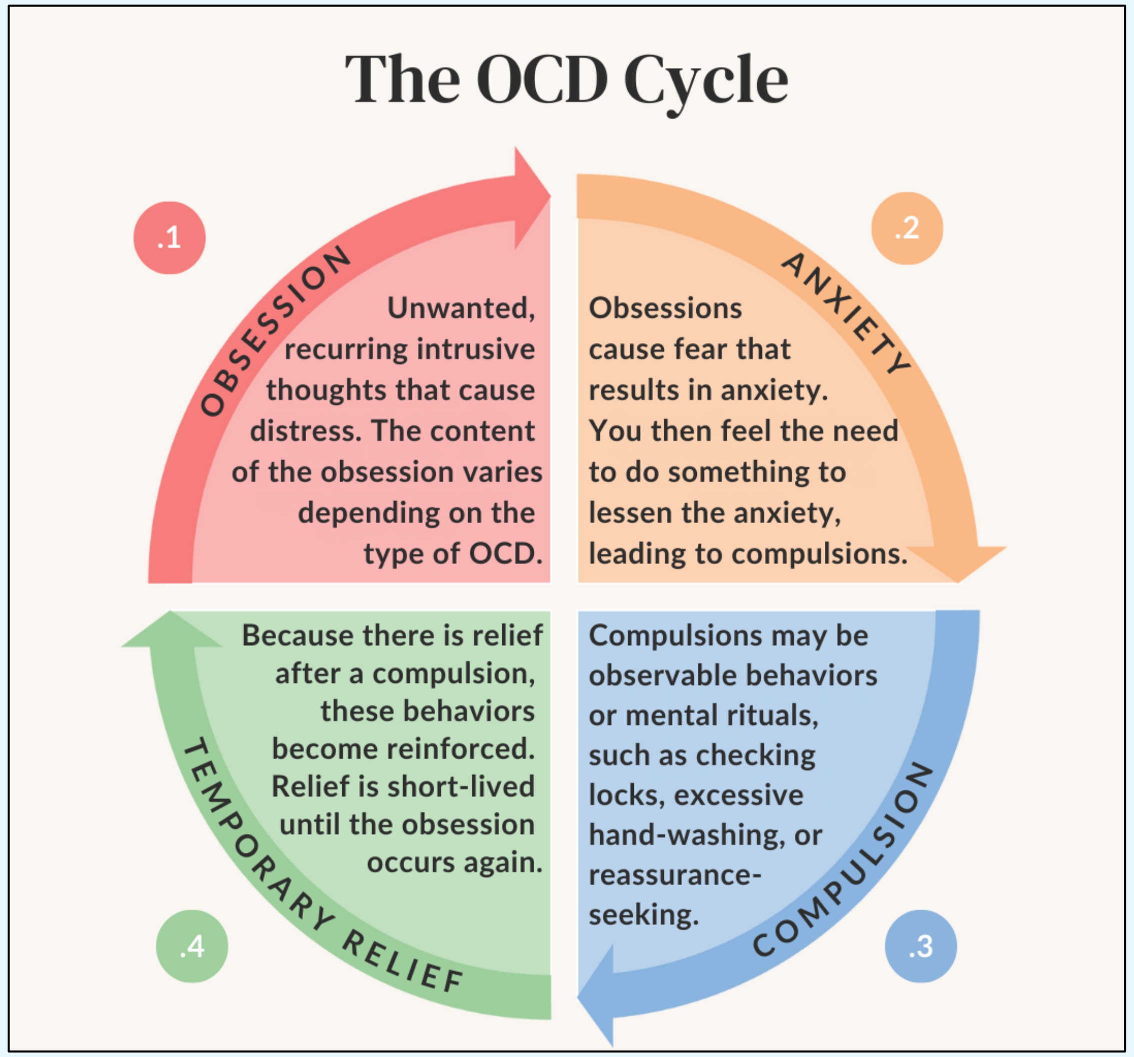
OCD professionals recommended a range of tools from symptom tracking to compulsion prevention to help clients manage their OCD outside of therapy. Dedicated OCD symptom tracking apps […] were suggested to easily record compulsions and exposure practices. They also recommended meditation apps […] for clients to practice mindfulness and listen to the bilateral stimulation playlist to calm the nervous system. These tools are intended for consistent out-of-session use to monitor symptoms and reinforce skills learned from therapy. **Professionals also recommended apps for in-situ use in response to triggered OCD episodes.**

[…]

Given the varied forms of triggers and compulsions (e.g., direct trigger interaction, consequence mitigation), **multimodal AI models can be used together with wearable devices** […] to identify and track triggers and compulsions beyond the scope of vision-based methods. Despite the merit of AI-based symptom-tracking automation, false-positive prediction results can potentially exacerbate OCD symptoms, since people with OCD often have low confidence in their memories [100]. To minimize this risk, **prediction models should communicate potential errors and involve users** in assessing the decision-making process using explainable AI techniques."

Source: *"It was Mentally Painful to Try and Stop": Design Opportunities for Just-in-Time Interventions for People with Obsessive-Compulsive Disorder in the Real World (2025)*

The OCD Enigma And the Impediment of ERP



- **Obsessive-Compulsive Disorder (OCD)** is a mental health condition characterized by intrusive thoughts (obsessions) and repetitive behaviours (compulsions) that significantly interfere with daily life.

- **Exposure Response Prevention (ERP) therapy** demonstrates the **most efficacy** in combating OCD**,** yet it remains very challenging to successfully apply it during **real-time compulsive surges** when conscious control is compromised.

- During the **early stages of a compulsive urge**, anxiety escalates rapidly, making it **increasingly difficult** for individuals to apply coping strategies **effectively**.

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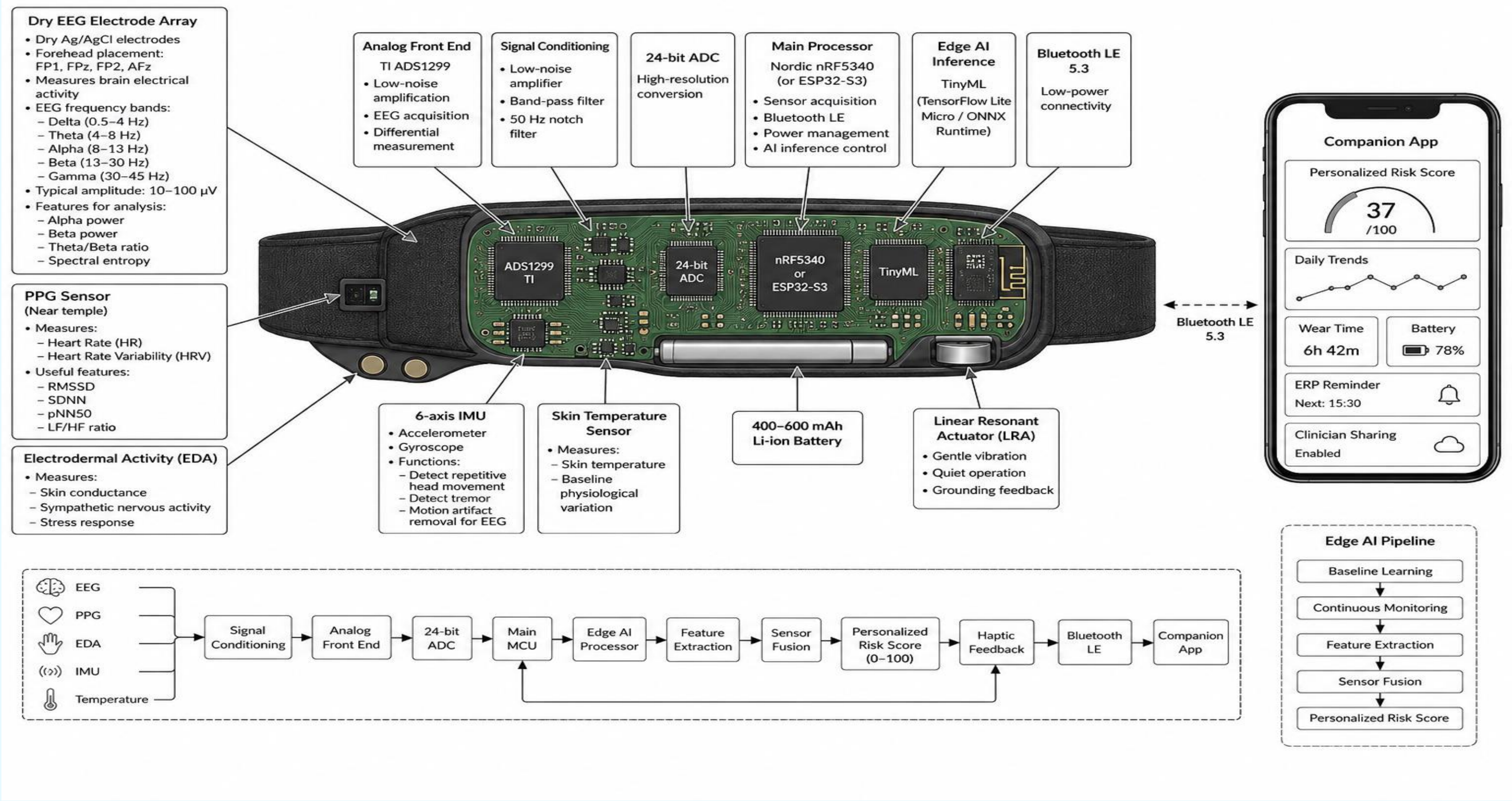
This critical intervention window—"Moment Zero"—remains largely unaddressed by existing wearable technologies.

Meet NeuroKavach!



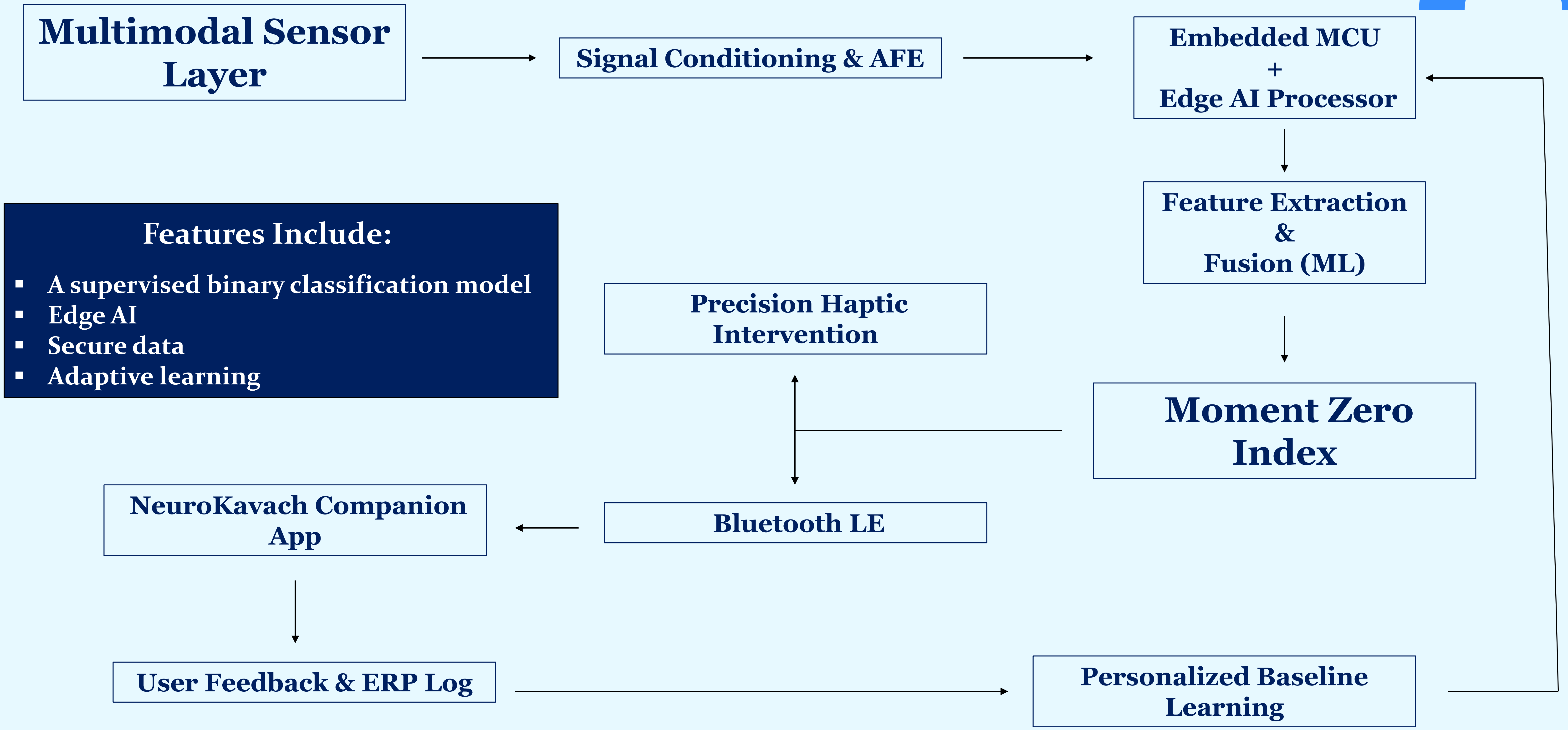
DATASETS USED

1. NinaPro DB1
2. NinaPro DB2
3. CapgMyo DB-a

HOW NEUROKAVACH WORKS

Physiological Signals → Embedded AI → 'Moment Zero' Detection → Haptic Cue → User Regains Control

Proposed System Architecture



Features Include:

- A supervised binary classification model
- Edge AI
- Secure data
- Adaptive learning

Design Challenges & Breakthroughs

Challenges	How We Addressed It
Lack of Unified OCD Physiological Datasets	Standardized preprocessing and common feature alignment across multiple datasets
High-Dimensional Multimodal Data	Feature selection and sensor fusion to reduce redundancy and improve model robustness
Personal Variability	Personalized baseline learning rather than one-size-fits-all thresholds
Real-Time Inference on Wearables	Lightweight Edge AI models optimized for low-power embedded hardware
False Positives & User Trust	Explainable AI with user feedback to continuously refine the personalized model

Why NeuroKavach?

- Personalized A.I.
- Moment Zero Intervention
- ERP Support
- Adaptive Learning
- Non-invasive Wearable

Results And Outcomes

Three EMG Datasets → Filtering → Normalization → Window Segmentation → Feature Extraction → Feature Selection → Machine Learning Model → Stable / Moment Zero Prediction

Our current prototype uses a supervised binary classification model for early-stage EMG signal classification. After preprocessing, feature extraction, and initial training, the model achieved **an accuracy of 80.23%**. This demonstrates that the proposed pipeline can distinguish between Stable and Moment Zero states with encouraging early-stage performance."

The Current Competitive Landscape

Feature	Meditation Apps	Smartwatch Stress	Therapy Alone	NeuroKavach
Real-Time Neuromuscular Detection	✗	✗	✗	✓
ERP-Specific Support	✗	✗	✓	✓
On-Device AI	✗	✗	✗	✓
Haptic Grounding	✗	Limited	✗	✓
Human-in-the-Loop Learning	✗	✗	✗	✓

NeuroKavach is the first assistive wearable targeting the "Moment Zero" of compulsive episodes.

MEET TEAM ZEROSTACK

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THANK YOU